SIMATS ENGINEERING

CSA3714- Software Testing

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Academic Integrity & Student Certification

I hereby certify that this expanded submission represents my original analytical work for assessment CSA3714 (CO2) and adheres fully to all academic integrity guidelines.

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1. Online Banking System – Fund Transfer Boundary Failures

Exhaustive Scenario Analysis & Failure Root-Cause

Repeated failures at minimum and maximum transfer thresholds in online banking stem from classic off-by-one comparison defects (e.g., using '>' instead of '>=') and improper interaction handling between transfer limits, daily cumulative allowances, and account balances. While random testing introduces broad exploratory variance, it fails to evaluate precise relational boundary conditions.

Equivalence Class Partitioning (ECP) & Boundary Value Analysis (BVA)

Assuming a standard transfer range of ₹100 to ₹50,000 per transaction:

- **EC1:** Valid Equivalence Class (VEC): Transfer amounts in [₹100, ₹50,000].
- **EC2:** Invalid Equivalence Class Below Minimum (IEC1): Transfer amounts < ₹100.
- **EC3:** Invalid Equivalence Class Above Maximum (IEC2): Transfer amounts > ₹50,000.
- **EC4:** Invalid Input Formats (IEC3): Negative values, zero, non-numeric strings, special characters, blank fields.

BVA requires testing boundary points (min-1, min, min+1, max-1, max, max+1):

Test ID	Boundary Point	Input Amount	Preconditions	Expected Result	Validation Layer
B1	Min - 1	₹99.00	Sufficient Balance	Reject: Below minimum limit error	UI & API Validation
B2	Min (Exact)	₹100.00	Sufficient Balance	Accept: Process transfer successfully	DB Ledger Debit/Credit
B3	Min + 1	₹101.00	Sufficient Balance	Accept: Process transfer	DB Ledger Debit/Credit

				successfully	
B4	Max - 1	₹49,999.00	Sufficient Balance	Accept: Process transfer successfully	DB Ledger Debit/Credit
B5	Max (Exact)	₹50,000.00	Sufficient Balance	Accept: Process transfer successfully	DB Ledger Debit/Credit
B6	Max + 1	₹50,001.00	Sufficient Balance	Reject: Exceeds single transfer limit	UI & API Validation
B7	Zero / Negative	₹0.00 / -₹10.00	Account Active	Reject: Invalid amount format	API Gateway Guard
B8	Format Error	ABC / Blank	Account Active	Reject: Non-numeric payload blocked	Front-end / API Schema

Critical Assessment of Random Testing vs. Systematic Strategy

Random testing randomly samples the input space, meaning the probability of hitting precise edge values like ₹100.00 or ₹50,000.00 is near zero in continuous float domains. Systematic boundary testing guarantees exact edge coverage. Random testing should only serve as a secondary fuzzing layer using deterministic seeds to uncover unexpected thread races or state corruptions.

Execution Plan & Verification Criteria

- **Execution Sequence:** Step 1: Execute 2-value and 3-value BVA scripts across API endpoints before UI rendering.

- **Multi-Condition Checks:** Step 2: Cross-validate amount boundaries against secondary business rules (e.g., balance = ₹100 but transfer fee = ₹5).
- **Ledger Verification:** Step 3: Database ledger assertion—rejected transfers MUST leave double-entry accounting ledgers untouched (zero net change); accepted transfers MUST enforce atomic transaction commits.

2. Healthcare Application – Inadequate White-Box Coverage

Exhaustive Scenario Analysis & Coverage Flaws

Deploying a healthcare application with critical post-release defects despite reaching 'high statement coverage' highlights the structural flaw of relying solely on statement coverage. Statement coverage only confirms that a line of code was touched—it does not evaluate unexecuted false branches, compound condition masking, or missing safety exception paths.

Formal Control Flow Graph (CFG) Representation

Consider a critical dosage assessment function with logic: Validate Patient -> Check Allergy -> Check Dosage Range -> Issue Alert/Approve.

CFG Structure & Paths:

- **N1:** Node 1: START -> Receive (Patient_ID, Drug_ID, Dose)
- **N2:** Node 2: Decision 1 (Is Patient Data Valid?) [Edges: N2->N3 (Yes), N2->N8 (No)]
- **N3:** Node 3: Decision 2 (Is Patient Allergic to Drug?) [Edges: N3->N4 (Yes), N3->N5 (No)]
- **N4:** Node 4: Action: Raise Critical Allergy Alert & Halt Order -> Edge to N8
- **N5:** Node 5: Decision 3 (Is Dose within Safe Range?) [Edges: N5->N6 (Yes), N5->N7 (No)]
- **N6:** Node 6: Action: Approve Prescription & Save Audit Log -> Edge to N8
- **N7:** Node 7: Action: Raise Overdose/Underdose Warning -> Edge to N8
- **N8:** Node 8: END

Branch vs. Statement vs. Path Coverage Analysis

Statement coverage requires executing N1, N2, N3, N4, N5, N6, N7, N8 at least once, which can be achieved in just 2 test cases. However, branch coverage mandates evaluating ALL decision edges (True and False for N2, N3, N5). Path coverage evaluates all combinations of branches (2^3 = 8 paths). Branch and condition coverage are indispensable in medical software to prevent untested safety branches.

Rigor Testing Implementation & Traceability

Metric	Target Minimum	Test Technique	Clinical Governance Role
Statement Coverage	100%	Basic Execution	Ensures no dead/unreachable code exists
Branch Coverage	100%	Decision Path Testing	Guarantees all safety checks have explicit decision evaluation
Condition Coverage	100%	MC/DC Analysis	Verifies independent condition influence on critical alerts
Mutation Coverage	> 90%	Fault Injection / Mutation	Validates that tests fail when logical operators are mutated
Requirements Traceability	100% Mapping	Traceability Matrix	Links every safety requirement (ISO 14971) to test code

3. Retail E-Commerce – Price, Quantity, and Discount Combinations

Scenario Analysis & Multi-Variable Interaction Risks

Testing price, quantity, and discount variables in isolation fails to catch complex multi-variable interactions. For example, a valid high quantity multiplied by a valid max price can cause numeric integer overflow, or a 100% discount combined with a non-zero quantity can produce zero or negative order totals when rounding rules interact with tax calculations.

Equivalence Partitioning & Decision Table Model

Variables: Price (P), Quantity (Q), Discount (D).

Rule ID	Price (P)	Quantity (Q)	Discount (D)	Expected System Behavior	Calculation Oracle Formula
R1	P <= 0 (Invalid)	Q > 0 (Valid)	D Valid (0-50%)	Reject Order: Invalid Price	N/A - Error Raised
R2	P Valid (>0)	Q <= 0 (Invalid)	D Valid (0-50%)	Reject Order: Invalid Quantity	N/A - Error Raised
R3	P Valid (>0)	Q Valid (>0)	D < 0% or > 100%	Reject Order: Invalid Discount	N/A - Error Raised
R4	P Valid ($100)	Q Valid (2)	D = 0%	Accept: Subtotal = $200	Total = P * Q

R5	P Valid ($100)	Q Valid (5)	D = 20%	Accept: Subtotal = $400	Total = (P * Q) * (1 - D)
R6	P Valid ($100)	Q Max (1000)	D Max (100%)	Accept: Total = $0.00	Capped at $0 min floor
R7	P Max ($99k)	Q Max (1000)	D = 10%	Accept: Verify No Overflow	64-bit BigDec precision

Combinatorial Optimization (Pairwise & Error Guessing)

When exhaustive 3-way Cartesian testing creates excessive test sets across multiple product categories, Pairwise (All-Pairs) testing reduces N^3 tests to N^2 while maintaining 100% 2-way interaction coverage. Error guessing targets specific edge scenarios: zero-price promo items, fractional quantities, floating-point rounding errors (e.g., $0.0000001 balances), and coupon code race conditions.

4. Complex Nested Conditions and Loops – Partial Path Coverage

Structural Analysis of Nested Control Structures

Nested conditions inside iterative loops lead to exponential path growth. Partial path coverage occurs when testing executes some loop iterations without exploring all internal branch paths or loop boundary conditions (0, 1, max iterations).

Mathematical Cyclomatic Complexity & CFG Derivation

Cyclomatic Complexity V(G) for a graph with E edges, N nodes, and P connected components:

Formula: $V(G) = E - N + 2P$ or $V(G) = \text{Predicate Nodes} + 1$

Consider an illustrative algorithm with 1 outer loop and 2 nested inner decisions:

- **Graph Metrics:** Nodes (N) = 7, Edges (E) = 9, Connected Components (P) = 1.

- **Complexity Calculation:** V(G) = 9 - 7 + 2(1) = 4 Basis Paths.

Independent Basis Paths Set:

- **Basis Path 1:** Path 1 (Zero Loop Iterations): N1 -> N2 (Loop Condition False) -> N7 (Exit)
- **Basis Path 2:** Path 2 (Loop 1st Pass, Inner Dec 1 True): N1 -> N2 -> N3 -> N4 -> N6 -> N2 -> N7
- **Basis Path 3:** Path 3 (Loop 1st Pass, Inner Dec 1 False, Dec 2 True): N1 -> N2 -> N3 -> N5 -> N6 -> N2 -> N7
- **Basis Path 4:** Path 4 (Loop 1st Pass, Inner Dec 1 False, Dec 2 False): N1 -> N2 -> N3 -> N6 -> N2 -> N7

Loop Boundary & Path Testing Rules

Loop Test Category	Iteration Count (n)	Purpose & Targeted Defect
Skip Loop	n = 0	Verifies initialization, empty data array handling, direct bypass
Single Pass	n = 1	Verifies initial loop body processing and state update
Double Pass	n = 2	Verifies loop re-initialization and cursor incrementation
Typical Pass	n = k (e.g., 50)	Verifies operational stability over multiple records
Boundary Pass	n = Max - 1, Max	Verifies upper buffer boundaries and off-by-one exit logic
Exceeded Pass	n = Max + 1	Verifies proper array index out-of-bounds

		protection

5. Financial Application – Integrated Black-Box and White-Box Testing

Critical Evaluation of Dual-Method Test Failures

Financial software that experiences production escapes despite passing separate black-box and white-box suites suffers from 'verification disconnect'. Black-box testing verifies functional outputs without ensuring code-path execution; white-box testing validates code paths without verifying persistent database state, ledger reconciliation, or business compliance.

Hybrid Integrated Test Framework Matrix

Transaction Type	Black-Box Test Input (ECP/BVA)	White-Box Control Flow Path	Database / Ledger Assertion
Fund Transfer	Amount = ₹0.00 (Boundary)	Input Guard Branch -> Early Return Error	Zero DB changes; Ledger unmodified
Overdraft Limit	Balance = ₹1,000; Transfer = ₹1,500	Balance Check -> Overdraft Branch -> Fee Calc	Audit trail entry logged; Overdraft flag set
Interest Posting	Accrual Date = Leap Day (Feb 29)	Date Branch -> Leap Year Formula Path	Interest ledger updated with exact 366-day factor
Concurrent Pay	2x Simultaneous ₹500 Debits	Thread Lock Path -> Idempotency Check	Exactly one debit processed; 2nd request

		Branch	rejected

Financial Data Integrity & Exit Criteria Framework

- **Ledger Balance Assertion:** Exit Criterion 1: 100% Requirements Traceability with verified double-entry ledger balance assertion.
- **Code Path Completeness:** Exit Criterion 2: 100% Branch and Condition Coverage on all transaction engine routines.
- **Resilience & Rollback:** Exit Criterion 3: Zero unhandled transaction rollback exceptions during simulated database connection drops.

6. HR Payroll – Bonus Errors at 5, 10, and 15 Years

Exhaustive Failure Analysis & Date-Derivation Defects

Bonus miscalculations occurring precisely at 5, 10, and 15-year anniversaries represent classic boundary operator defects (using '>' instead of '>=') and improper date math handling (e.g., leap years, timezones, or calculating tenure based on calendar years versus exact anniversary timestamps).

Boundary Value Analysis (BVA) & Equivalence Partitioning

Service Tenure Bands: Band 0 (<5 yrs: 0%), Band 1 (5 to <10 yrs: 5%), Band 2 (10 to <15 yrs: 10%), Band 3 (>=15 yrs: 15%).

Test ID	Tenure Input (Exact Date Offset)	Target Service Band	Expected Bonus %	Validation Focus
TB1	4 Years, 364 Days	Band 0 (<5 Yrs)	0%	Lower Boundary - 1 Day
TB2	Exactly 5 Years (0	Band 1 (5-10 Yrs)	5%	Exact Threshold

	Days)			(>= 5 Check)
TB3	5 Years, 1 Day	Band 1 (5-10 Yrs)	5%	Upper Boundary + 1 Day
TB4	9 Years, 364 Days	Band 1 (5-10 Yrs)	5%	Lower Boundary - 1 Day
TB5	Exactly 10 Years (0 Days)	Band 2 (10-15 Yrs)	10%	Exact Threshold (>= 10 Check)
TB6	10 Years, 1 Day	Band 2 (10-15 Yrs)	10%	Upper Boundary + 1 Day
TB7	14 Years, 364 Days	Band 2 (10-15 Yrs)	10%	Lower Boundary - 1 Day
TB8	Exactly 15 Years (0 Days)	Band 3 (>=15 Yrs)	15%	Exact Threshold (>= 15 Check)

Decision Table: Service Band x Performance Rating

Crossing tenure bands with Performance Ratings (Exceeds = +2%, Meets = +0%, Needs Improvement = Bonus Forfeited):

- **Rule 1:** Tenure = 5 Yrs (Band 1: 5%) AND Rating = 'Exceeds' -> Total Bonus = 7%.
- **Rule 2:** Tenure = 5 Yrs (Band 1: 5%) AND Rating = 'Needs Improvement' -> Total Bonus = 0% (Forfeited).

7. Laboratory Temperature Controller and MC/DC Coverage

Safety Analysis & Limitations of Branch Coverage

In life-critical or laboratory environments, branch coverage alone is insufficient. A decision with compound Boolean logic can evaluate to True and False overall without individual atomic conditions ever independently determining the outcome, masking dangerous hardware fault states.

Boolean Decision Model & Formal MC/DC Truth Table

Let Alarm Trigger Decision D = H OR L OR S

- **Condition H:** H = High Temp Exceeded (Temp > 80°C)
- **Condition L:** L = Low Temp Exceeded (Temp < 0°C)
- **Condition S:** S = Sensor Malfunction Detected

Modified Condition / Decision Coverage (MC/DC) Test Set:

Test Run	Condition H	Condition L	Condition S	Decision Outcome (D)	MC/DC Independence Demonstration
T1	False (0)	False (0)	False (0)	False (0)	Base Baseline Case
T2	True (1)	False (0)	False (0)	True (1)	T1 vs T2 proves Condition H independently affects D
T3	False (0)	True (1)	False (0)	True (1)	T1 vs T3 proves Condition L independently affects D
T4	False (0)	False (0)	True (1)	True (1)	T1 vs T4 proves Condition S independently affects D

White-Box Integration & Fault Injection Strategy

- **Fault Injection:** Hardware-in-the-Loop (HIL) testing to inject simulated open-circuit voltage (Sensor Fault = True).
- **Actuator Verification:** Verify fail-safe hardware actuator overrides trigger instantly upon alarm activation.

8. Travel Promo-Code Combinations and Negative Totals

Root-Cause Analysis of Stacked Promotion Failures

Negative invoice totals occur when multiple promotion codes are applied without order-of-precedence validation, discount capping rules, or absolute price floor checks. Stacked fixed discounts ($50 off + $60 off on a $100 fare) result in negative balances if applied directly.

Comprehensive Decision Table for Promo Stacking

Rule	Base Price	Code A (Fixed $50)	Code B (% Discount)	User Tier	Expected Payable Total	System Defense Guard
R1	$40.00	Applied ($50)	None	Standard	$0.00	Discount capped at Base Price ($40 max discount)
R2	$100.00	Applied ($50)	Applied (20%)	VIP	$40.00	Order of precedence: % discount applied first, then fixed
R3	$100.00	Incompatible Code	Incompatible Code	Standard	Reject 2nd Code	Mutually exclusive code combination guard

						triggered
R4	$30.00	Applied ($50)	Applied (50%)	VIP	$0.00	Absolute floor rule: Payable total CANNOT be < $0.00

Cause-Effect Graphing & Validation Oracles

Causes (Valid Fare, Eligible Tier, Active Code) are mapped to Effects (Apply Flat, Apply %, Cap Floor). The payment gateway payload must be asserted against the server-side ledger calculation to eliminate client-side tamper risks.

9. Database Migration – Loops, Infinite Loops, and Cyclomatic Complexity

Why Statement Coverage Fails to Detect Non-Termination

High statement coverage merely confirms that loop body statements executed at least once during a test run. It fails to detect infinite loop bugs caused by unadvanced database cursors, missing loop variable increments on error-handling branches, or malformed data records.

Control Flow Graph & Mathematical Complexity

Migration Logic: Fetch Record -> Validate -> Transform -> Write DB -> Advance Cursor -> Check EOF.

V(G) = E - N + 2P = 8 Edges - 6 Nodes + 2(1) = 4 Basis Paths.

Loop Testing Strategy & Termination Safeguards

Test Case Type	Input Data Composition	Targeted Defect / Infinite Loop Guard
Empty Source	0 Records in Table	Validates zero-iteration loop bypass and clean exit
Single Record	1 Valid Record	Validates cursor start and immediate EOF termination
Malformed Record	Record with NULL Primary Key	Verifies error path still advances cursor to avoid infinite loop
Large Batch	100,000 Records	Validates memory retention and cursor buffer rollover
Timeout Guard	Locked Database Row	Verifies watchdog timer breaks infinite lock waiting

10. Food Delivery App – State Transition Testing

Failure Analysis of Workflow State Tracking

Orders becoming 'stuck' in Out for Delivery upon driver cancellation indicate missing or unhandled state transitions in the dynamic order lifecycle state machine. Inputs alone (ECP/BVA) cannot test state sequence validity.

Formal State Transition Matrix & Lifecycle Model

Current State	Event / Trigger	Next Valid State	System Action & Handler
Order Placed	Restaurant Accepts	Preparing Food	Send push notification; assign kitchen order
Preparing Food	Food Ready	Driver Assigned	Notify nearby drivers; update ETA
Driver Assigned	Driver Picked Up	Out for Delivery	Enable live GPS tracking
Out for Delivery	Driver Cancels Order	Reassignment Pending	Alert dispatch service; re-route to new driver
Reassignment Pending	New Driver Accepts	Out for Delivery	Update tracking UI with new driver details
Out for Delivery	Customer Confirms	Delivered	Terminal state; trigger payment payout
Delivered	Driver Cancels Event	Delivered (No-Op)	Reject illegal transition attempt

State Transition Coverage Criteria

- **Transition Pair Coverage:** 1-Switch Coverage: Tests all valid pairs of consecutive transitions (e.g., Out for Delivery -> Driver Cancels -> Reassignment Pending -> New Driver Accepts).

- **Illegal Event Handling:** Negative State Testing: Attempting invalid transitions (e.g., Order Placed -> Delivered) must throw explicit StateTransitionExceptions.

11. Cybersecurity Multi-Factor Authentication – Hybrid Testing

Security Assessment of Brute-Force Vulnerabilities

Brute-force and rate-limiting vulnerabilities escape detection when testing focuses solely on isolated valid/invalid authentication attempts rather than stateful attack patterns, parallel request bursts, and OTP expiry windows.

White-Box Path Analysis & Security Controls

Attack Scenario / Path	Black-Box Error Guessing Technique	White-Box Control Flow Guard	Expected Protection Outcome
Password Brute Force	Send 10 invalid passwords rapidly	Counter Branch >= 5 Attempts -> Lock	Account locked for 15 mins; HTTP 429 Too Many Requests
OTP Replay Attack	Submit intercepted OTP twice	OTP DB Flag Check -> Used Path	Reject second attempt; invalidate OTP session
OTP Expiry Bypass	Submit OTP after 5 minutes	Timestamp Comparison Branch	Reject expired OTP; prompt for fresh request
Concurrent OTP Race	Send 50 parallel requests with 1 OTP	Database Row Locking / Mutex	Exactly 1 request succeeds; 49 requests fail gracefully

12. E-Learning Subscription Expiry and Upgrades

Failure Analysis of Timezone & Date-Boundary Logic

Subscription upgrade failures occurring on the exact expiry day stem from ambiguous timestamp definitions (e.g., midnight UTC vs local timezone) and improper entitlement evaluation on boundaries.

Equivalence Partitioning & Precision Date Boundary Matrix

Test Case	Subscription Expiry Time	Upgrade Event Timestamp	Expected Access Entitlement
Pre-Expiry Upgrade	2026-11-15 23:59:59 UTC	2026-11-14 10:00:00 UTC	Seamless transition to Premium; prorate refund
Exact Expiry Instant	2026-11-15 23:59:59 UTC	2026-11-15 23:59:59 UTC	Upgrade processed without service interruption
Post-Expiry Upgrade	2026-11-15 23:59:59 UTC	2026-11-16 00:00:01 UTC	Basic plan expired; fresh purchase of Premium plan
Timezone Shift	2026-11-15 23:59:59 EST	2026-11-16 02:00:00 UTC	Evaluated against normalized UTC timestamps

13. Smart-Home Lighting – Branch and Condition Coverage

White-Box Structural Evaluation of Firmware Logic

Firmware logic relying on complex switch statements and compound ambient light conditions can experience stuck-state errors if switch cases miss default fallbacks or boolean conditions mask manual override commands.

Decision & Condition Truth Matrix

Decision: Light ON = (Ambient Light < 50 Lux AND Motion Detected = True) OR Manual Override = True

Test Case	Ambient Light < 50 Lux	Motion Detected	Manual Override	Light State	Branch Evaluated
T1	False (100 Lux)	False	False	OFF	Automatic OFF Path
T2	True (20 Lux)	True	False	ON	Automatic ON Path
T3	False (100 Lux)	False	True	ON	Manual Override Path
T4	Sensor Fault (ERR)	N/A	False	OFF (Fail-Safe)	Default Switch Case Path

14. Shipping-Cost Robustness Testing

Robust Boundary Value Analysis (7-Point Testing)

Missing zero and negative weight checks exposes logistics engines to erroneous quote calculations or free shipping exploits. Robust BVA evaluates: Min-, Min, Min+, Nom, Max-, Max, Max+.

Robust BVA Test Suite (Weight Range: 0.01 kg to 50.00 kg)

BVA Point	Weight Input	Expected System Reaction	Validation Guard
Min - 1 (Invalid)	-0.01 kg	Reject: Negative weight invalid	Input Validation Filter
Min Boundary (Zero)	0.00 kg	Reject: Zero weight invalid	Input Validation Filter
Min + 1 (Valid Min)	0.01 kg	Accept: Calculate base tariff	Billing Engine
Nominal Value	25.00 kg	Accept: Calculate standard tariff	Billing Engine
Max - 1	49.99 kg	Accept: Calculate max standard tariff	Billing Engine
Max Boundary	50.00 kg	Accept: Calculate max standard tariff	Billing Engine
Max + 1 (Exceeded)	50.01 kg	Reject: Exceeds standard parcel limit	Freight Handler Routing

15. Medical Imaging Parallel Processing & Prime Data Sizes

Root-Cause Analysis of Parallel Partitioning Defects

Failures occurring specifically with prime-sized datasets (e.g., 101 pixels or prime image dimensions) indicate improper array chunking, missing remainder collection nodes, or worker thread synchronization race conditions.

Basis Path & Concurrency Test Matrix

CFG Nodes: N1 (Receive Array) -> N2 (Calculate Chunk Size) -> N3 (Has Remainder?) -> N4 (Spawn Workers) -> N5 (Assign Remainder to Final Worker) -> N6 (Aggregate Results).

Dataset Size	Worker Threads	Data Characteristic	Expected Partition Distribution	Validation Oracle
100 Elements	4 Threads	Composite (Even)	25, 25, 25, 25 elements	Serial vs Parallel Pixel Output Match
101 Elements	4 Threads	Prime Size	25, 25, 25, 26 (Remainder handled)	Serial vs Parallel Pixel Output Match
1 Element	4 Threads	Boundary Min	1 worker gets 1; 3 idle gracefully	No IndexOutOfBounds Exception
0 Elements	4 Threads	Invalid Input	Reject immediately	Return empty array / validation error